

Canonical Residual Erasure and the Native Affine Gram at a Fixed Shift: Exact Alias Defects, Dark Spaces, and Two-Stage Transfer Gates

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Abstract

The low-sign average and the physical all-minus corner of a large-prime Walsh packet produce two different Gram forms. The former resolves the low residual label s ; the latter erases s inside each native output coordinate. We identify the missing map exactly. On the canonical low-Walsh coefficient space it is the signed fiber sum

$$(\mathbf{E}_\mu \mathbf{z})_i = \sum_t \mu(t) z_{i,t},$$

or, after the physical Möbius gauge has already been absorbed, the unsigned erasure $(\mathbf{E}_1 \mathbf{A})_i = \sum_s A_{i,s}$. Its quadratic defect is the complete signed off-diagonal exchange form. Each active output fiber has one bright direction and a dark space of codimension one. Thus any support-only lower comparison fails as soon as two residual labels share a native output; this obstruction is sharp and is not removed by terminal-prime singleton incidence.

For the actual coefficient image $E = \text{Ran } T$, we prove that the optimal uniform native comparison is the least eigenvalue of the compression of $\mathbf{E}_\mu^* \mathbf{E}_\mu$ to E . We give its exact generalized-Gram and Moore–Penrose forms, a dimension kill test, and the necessary fixed-carrier convention. At one fixed nonzero shift h_0 , the terminal residual vector from the preceding prime-fiber construction is identified coordinate by coordinate with the gauged high-Walsh lift. The complete native affine column on the declared slice then factors through two independent gates: a residual-label alias constant and a base-high affine noncancellation constant. Neither constant is proved positive at the endpoint. The endpoint very-high face also lies in a smaller cofactor band than the inherited $R \asymp J$ packet, so its activation requires a separate relocation/occupancy theorem. The results are exact L0 linear algebra and L1 interface closure; they do not give a parity improvement or a twin-prime conclusion.

Keywords: residual-label erasure; native affine Gram; dark space; generalized Rayleigh quotient; fixed physical shift; parity boundary.

Convention. One nonzero physical shift h_0 , one direct residue/orbit slice, and one side of the inherited terminal system are fixed unless a left/right direct sum is displayed. All Hilbert norms use raw counting measure. The cutoff profile is written as $\Lambda_X(n) = (\log n) \Phi_0(n/X)$; this is the normalized version of the same shaped terminal weight denoted Λ in the inherited papers and is not the von Mangoldt function. Every label multiplicity is taken on a fixed baseline carrier after identically zero coordinates, but not coefficient-dependent cancellations, have been removed.

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1 Introduction

The fixed-shift residual program has reached a point at which two nonnegative quantities that look similar must be separated sharply. The very-high-prime decomposition of TPC-45 produces the physical all-minus affine corner

$$\mathcal{N}_{\text{nat}} := \left\| v_0 - \sum_{p>y} v_p \right\|^2, \quad (1)$$

whereas averaging the low Walsh signs produces

$$\mathcal{R}_{\text{low}} := \mathcal{D}_0 - \mathcal{D}_H + \left\| \sum_{p>y} x_p \right\|^2. \quad (2)$$

For each individual p , the diagonal norms agree, $\|v_p\| = \|x_p\|$, but the cross inner products do not [2]. The prime-resolved work in TPC-55–57 addresses the coherent sum over p at a fixed residual label. It does not yet return that residual energy to the native output space [4–6].

The missing operation is simple to state and consequential to analyze:

$$p \mapsto (i, s) \quad \text{followed by} \quad (i, s) \mapsto i. \quad (3)$$

The first arrow groups terminal primes while retaining the residual label s . The second erases s inside the native output coordinate i . TPC-57 studies the first arrow. This paper studies the second. In particular, one prime in every (i, s) -fiber does not imply one residual label in every i -fiber.

1.1 The exact erasure identity

Fix the baseline active residual labels $\mathcal{L}_i$ at output i , and write $z_i = (z_{i,t})_{t \in \mathcal{L}_i}$. Once the Möbius gauge is included, the native map is

$$(\mathbf{E}_\mu z)_i := \sum_{t \in \mathcal{L}_i} \mu(t) z_{i,t}. \quad (4)$$

The entire discrepancy between resolved and native energies is

$$\|\mathbf{E}_\mu z\|^2 - \|z\|^2 = 2 \operatorname{Re} \sum_i \sum_{t < t'} \mu(t) \overline{\mu(t')} z_{i,t} \overline{z_{i,t'}}. \quad (5)$$

For $k_i = |\mathcal{L}_i| > 0$, the i -block has one nonzero singular value $\sqrt{k_i}$ and a $k_i - 1$ dimensional kernel. The Möbius signs rotate this decomposition by a diagonal unitary; they do not eliminate the kernel. Hence a support-only lower inequality

$$\|\mathbf{E}_\mu z\|^2 \geq c \|z\|^2 \quad (6)$$

with $c > 0$ is false on the full ambient space whenever any $k_i \geq 2$. This negative result is exact, but its quantifier is equally important: it does not show failure on the smaller image traced out by the literal physical coefficients.

1.2 The actual coefficient image

Let a fixed baseline carrier be chosen independently of the bounded row coefficient vector ζ , and let

$$T : \mathbb{C}^{\mathcal{M}} \longrightarrow \mathcal{H}_W, \quad z = T\zeta, \quad E := \operatorname{Ran} T. \quad (7)$$

The best coefficient-uniform comparison is

$$\alpha_E := \inf_{0 \neq z \in E} \frac{\|\mathbf{E}_\mu z\|^2}{\|z\|^2}. \quad (8)$$

In finite dimension it is the least eigenvalue, including zero, of the compression of $\mathbf{E}_\mu^* \mathbf{E}_\mu$ to E , and

$$\alpha_E > 0 \iff E \cap \text{Ker } \mathbf{E}_\mu = \{0\}. \quad (9)$$

With

$$R := T^* T, \quad N := T^* \mathbf{E}_\mu^* \mathbf{E}_\mu T, \quad \mathcal{S} := \text{Ran } R, \quad (10)$$

the exact relative defect is

$$\Theta := R^{\dagger/2} (N - R) R^{\dagger/2} \big|_{\mathcal{S}}, \quad \alpha_E = 1 + \lambda_{\min}(\Theta). \quad (11)$$

The restriction to $\mathcal{S}$ is essential: zero directions in $\text{Ker } T$ must not be counted as physical generalized eigenvectors.

There are therefore three different questions:

- (i) the ambient question, which has a sharp dark-space obstruction;
- (ii) the coefficient-uniform question on the actual image E ; and
- (iii) the distinguished-vector question for the one literal coefficient vector inherited at scale X .

Failure of (i) does not imply failure of (ii) or (iii). Conversely, a numerically favorable distinguished vector does not prove a uniform bound on E .

1.3 The fixed-shift two-stage gate

Use the sign convention

$$z_H(i, s) := - \sum_{p > y} b_{i, sp}, \quad h := \mathbf{E}_{\mu, H} z_H = - \sum_{p > y} v_p. \quad (12)$$

The TPC-57 terminal residual amplitude already contains the common physical Möbius sign $\mu(s)$. If that vector is denoted by A , then the literal interface is

$$A = \mathbf{U}_\mu z_H, \quad \mathbf{E}_1 A = h, \quad \|A\| = \|z_H\|. \quad (13)$$

Thus no second Möbius factor is inserted when the TPC-57 output is grouped natively.

For the actual high coefficient image define α_H by the analogue of (8). Separately, on the actual paired image of v_0 and h , define

$$\delta_{\text{aff}} := \inf_{\substack{(b, g) \in \mathcal{F}_{\text{aff}} \\ (b, g) \neq (0, 0)}} \frac{\|b + g\|^2}{\|b\|^2 + \|g\|^2}. \quad (14)$$

Here $\mathcal{F}_{\text{aff}}$ is the nonzero image of the coupled map driven by the same row vector in both entries, defined precisely in [section 6](#). Then the exact route is

$$\mathcal{N}_{\text{nat}} = \|v_0 + h\|^2 \geq \delta_{\text{aff}} \alpha_H \|z_H\|^2. \quad (15)$$

The constants are independent gates. Residual labels may cancel under native erasure even if the terminal-prime fibers are singletons, and the base vector may cancel the high vector even if native high energy is positive.

There is also an upstream support gate. At the endpoint cutoff $y = X^{267/400 + \eta}$ with fixed $\eta > 0$, the relation $pr = mj + h_0 \leq X^{1+o(1)}$ forces $r \leq X^{133/400 - \eta + o(1)}$. Hence this very-high face is eventually disjoint from the older dyadic block $R \asymp J = X^{133/400 + o(1)}$. The interface identities remain exact on their declared face, but they do not prove that the face carries a positive fraction of the inherited energy. A compatible-band relocation and occupancy estimate is therefore charged separately in the endpoint ledger.

1.4 Claim boundary and organization

The finite-dimensional erasure spectrum, compressed Rayleigh quotient, generalized-Gram formula, and counterexamples are unconditional L0 results. The coordinate dictionary at one fixed $h_0 \neq 0$ is an L1 interface statement on the declared baseline carrier. A positive polynomial lower bound for either α_H or δ_{aff} , relocation to and occupancy of the compatible smaller cofactor band, terminal activation at the endpoint, boundary/tail reconnection, and an estimate for the original arithmetic defect remain open. In particular, no parity improvement and no twin-prime conclusion is asserted; the classical parity boundary remains in force [1].

Section 2 fixes the two Gram forms and their literal signs. Section 3 proves the exact erasure spectrum, and section 4 gives the actual-image and generalized-Gram theorems. The fixed-shift coordinate match is proved in section 5; the full affine completion follows in section 6. Sections 7 and 8 record the transport restrictions, endpoint budget, kill criteria, and next arithmetic gate.

2 Two inherited Gram forms and their common label dictionary

The comparison problem starts with two exact identities from TPC-45, not with two names for the same quadratic form. We restate the identities on one side of the equality column, retain the shaped terminal coefficient, and insert the two successive quotient maps that are needed later. The right side is obtained by interchanging the two source systems; the full column is their orthogonal direct sum. No estimate in this section mixes the two sides before that direct sum is taken.

2.1 The very-high cutoff and the baseline atoms

Fix the one physical intercept

$$h_0 \in \mathbb{Z} \setminus \{0\}. \quad (16)$$

Let $N_{\pm}$, $D_{\pm}$, J_{+} , and Δ_M be the support endpoints in the TPC-45 equality packet. The very-high cutoff $y = y_X$ is required to satisfy both inherited conditions

$$y > \max\{|h_0|, D_{+}, J_{+}, \Delta_M, \sqrt{N_{+}}\}, \quad (17)$$

$$\frac{D_{+}N_{+}}{y} < D_{-}N_{-}. \quad (18)$$

At the endpoint one may take $y = X^{267/400+\eta}$ for fixed $0 < \eta < 133/400$, but no exponent calculation below uses more than (17)–(18).

This endpoint choice changes the admissible cofactor band. On the physical equality support,

$$pr = mj + h_0 \leq N_{+} = X^{1+o(1)}, \quad p > y, \quad (19)$$

and therefore

$$\boxed{r \leq \frac{N_{+}}{y} = X^{133/400-\eta+o(1)}}. \quad (20)$$

For fixed $\eta > 0$, this is eventually disjoint from the inherited block $R \asymp J = X^{133/400+o(1)}$. Thus every identity below is exact on the declared very-high face, but applying it to endpoint energy requires a separate lower transfer into a compatible smaller r -band. The old $R \asymp J$ block may otherwise make the identities empty truths; no occupancy statement is inferred from the cutoff alone.

Write $i \in \mathcal{I}$ for one native output coordinate. On the left side, $i = (L, \gamma, j)$; on the right side, $i = (R, \alpha, j)$. TPC-45 associates to each actually occurring squarefree equality label u a stripped coefficient $b_{i,u} \in \mathbb{C}$. The word *stripped* means that the Möbius character $\mu(u)$ is not contained

in $b_{i,u}$; the literal stripped source amplitude, mask, smooth profiles, and shaped terminal weight are retained. We use the canonical active label set

$$\mathcal{U}_i^\circ := \{u : b_{i,u} \text{ is not identically zero on the declared baseline carrier}\}. \quad (21)$$

For one distinguished coefficient vector some entries on this fixed carrier may vanish. They remain zero coordinates; they do not change the definition of the carrier or create a new representation.

The cutoff conditions give the unique factorization

$$u = sq, \quad P^+(s) \leq y, \quad q = 1 \text{ or } q = p \in \mathcal{P}_y, \quad (22)$$

where $\mathcal{P}_y$ is the set of occurring primes $p > y$. If $q = p$, then $s = dr$ on the literal equality support. For each fixed pair (i, p) there is at most one source row on the chosen side. This last statement is the inherited fixed- (j, p) singleton incidence; it does not say that a fixed (i, s) contains at most one terminal prime.

2.2 The native affine vectors

In the native output Hilbert space

$$\mathcal{H}_{\text{nat}} := \ell^2(\mathcal{I}) \quad (23)$$

define

$$(v_0)_i := \sum_{\substack{u \in \mathcal{U}_i^\circ \\ u_{>y}=1}} \mu(u) b_{i,u}, \quad (24)$$

$$(v_p)_i := \sum_{s: sp \in \mathcal{U}_i^\circ} \mu(s) b_{i,sp}, \quad p \in \mathcal{P}_y. \quad (25)$$

The sum in (25) has at most one nonzero term for fixed (i, p) , but writing the sum keeps the output map visible. At the physical all-minus prime-sign point the exact native column is

$$\boxed{S_{\text{nat}}(h_0) = v_0 - \sum_{p \in \mathcal{P}_y} v_p}, \quad \mathcal{N}_{\text{nat}}(h_0) = \left\| v_0 - \sum_{p \in \mathcal{P}_y} v_p \right\|_{\mathcal{H}_{\text{nat}}}^2. \quad (26)$$

The minus sign is the physical value of the unique prime $p > y$. It is part of the affine identity and cannot be discarded when the base-high cross term is later formed [2].

2.3 The residual-label Gram and the Möbius gauge

The high residual Hilbert space is

$$\mathcal{H}_{\text{res}}^H := \ell^2\{(i, s) : i \in \mathcal{I}, P^+(s) \leq y, \text{ and } sp \in \mathcal{U}_i^\circ \text{ for some } p \in \mathcal{P}_y\}. \quad (27)$$

For $p \in \mathcal{P}_y$, the TPC-45 residual vector is the *stripped* vector

$$(x_p)_{i,s} := \begin{cases} b_{i,sp}, & sp \in \mathcal{U}_i^\circ, \\ 0, & sp \notin \mathcal{U}_i^\circ. \end{cases} \quad (28)$$

Define the residual Möbius gauge and the unsigned residual-label erasure by

$$(\mathbf{U}_\mu z)_{i,s} := \mu(s) z_{i,s}, \quad (29)$$

$$(\mathbf{E}z)_i := \sum_s z_{i,s}. \quad (30)$$

Squarefreeness makes $\mathbf{U}_\mu$ unitary. The definitions give the coordinatewise identity

$$\boxed{\mathbf{E}\mathbf{U}_\mu x_p = v_p.} \quad (31)$$

Consequently

$$\|x_p\|_{\mathcal{H}_{\text{res}}^H} = \|\mathbf{U}_\mu x_p\|_{\mathcal{H}_{\text{res}}^H} = \|v_p\|_{\mathcal{H}_{\text{nat}}}. \quad (32)$$

The last equality uses the fixed- (i, p) singleton incidence. It is not an assertion that $\mathbf{E}$ is an isometry on a sum of different prime columns.

To fix the inherited global sign once and for all, put

$$\boxed{z_H := - \sum_{p \in \mathcal{P}_y} x_p, \quad A_H := \mathbf{U}_\mu z_H.} \quad (33)$$

Thus z_H is the stripped resolved high vector, whereas A_H is its physical residual-Möbius version. Equations (31) and (33) imply

$$\boxed{\mathbf{E}A_H = - \sum_{p \in \mathcal{P}_y} v_p.} \quad (34)$$

This is the sign convention used throughout TPC-58.

The low-sign-averaged identity of TPC-45 is

$$\boxed{\mathbb{E}_{q \leq y} Z_0(\varepsilon_{\leq y}, -\mathbf{1}_{>y}) = \mathcal{D}_0 - \mathcal{D}_H + \|A_H\|_{\mathcal{H}_{\text{res}}^H}^2.} \quad (35)$$

Here $\mathcal{D}_0$ is the raw stripped atomic diagonal and $\mathcal{D}_H$ is its very-high-prime part. The equality $\|A_H\| = \|z_H\| = \|\sum_p x_p\|$ uses only the common diagonal gauge. In particular, the last term in (35) already contains the coherent cross terms between different terminal primes that share the same (i, s) . It is not the terminal atomic diagonal $\sum_p \|x_p\|^2$.

Proposition 2.1 (Exact two-Gram dictionary). *Under (17)–(18), the native and residual very-high faces are related by*

$$\sum_p \|v_p\|^2 = \sum_p \|x_p\|^2 = \mathcal{D}_H, \quad (36)$$

$$\left\| - \sum_p v_p \right\|^2 = \|\mathbf{E}A_H\|^2, \quad (37)$$

$$\left\| \sum_p x_p \right\|^2 = \|A_H\|^2. \quad (38)$$

For $p \neq q$, however,

$$\langle v_p, v_q \rangle_{\mathcal{H}_{\text{nat}}} = \sum_i \sum_{s,t} \mu(s) \mu(t) b_{i,sp} \overline{b_{i,tq}}, \quad (39)$$

$$\langle x_p, x_q \rangle_{\mathcal{H}_{\text{res}}^H} = \sum_i \sum_s b_{i,sp} \overline{b_{i,sq}}. \quad (40)$$

Thus the residual Gram retains only $s = t$, whereas the native Gram also contains every $s \neq t$ alias pair sharing the same output i .

Proof. The common diagonal is (32) summed over p . Equations (37) and (38) follow from (33)–(34). Expanding the two inner products gives (39)–(40). $\square$

The proposition is an exact L1 dictionary for the inherited face. It does not give either inequality between $\|\mathbf{E}A_H\|^2$ and $\|A_H\|^2$. Nor may the nonnegative base term $\mathcal{D}_0 - \mathcal{D}_H$ in (35) be replaced by $\|v_0\|^2$: the former is obtained after low-sign orthogonalization, while the latter retains the native base–base cross terms.

2.4 The two quotient maps are not interchangeable

For a literal terminal atom write its prime-resolved key as

$$(i, s, p), \quad i = (L, \gamma, j) \text{ or } (R, \alpha, j), \quad s = dr. \quad (41)$$

The two maps in the physical chain are

$$\boxed{(i, s, p) \longmapsto (i, s) \longmapsto i.} \quad (42)$$

The first arrow coherently sums terminal primes in a fixed residual label; it is the $p \rightarrow (i, s)$ grouping studied in TPC–55–57 [4–6]. The second arrow is the unsigned erasure $\mathbf{E}$; it aliases different residual labels s in the same native output coordinate. A singleton for the first arrow need not be a singleton for the second. In particular, a lower square proved after the first grouping is an input to the native-alias problem, not its solution.

All displayed sums use raw counting measure on their stated canonical coordinates. Prime-resolved injectivity is used before the first arrow, and the p -sum is formed exactly once in each (i, s) . Duplicating a prime-resolved atom, adjoining an ambient affine-prime solution, or inserting cancelling null pairs would change the Hilbert space and is not part of this dictionary.

3 Canonical residual erasure and its exact spectrum

This section isolates the finite-dimensional linear algebra of forgetting a low label after its Möbius phase has been restored. No arithmetic distribution statement is used. The distinction between a canonical coefficient array and its erased physical image is essential: the first is unique in its declared orthogonal basis, whereas the second generally has a large dark space.

3.1 The block erasure operator

Let $\mathcal{I}$ be a finite set. For each $i \in \mathcal{I}$, let $\mathcal{T}_i$ be a nonempty finite label set of size k_i , and put

$$\mathcal{Z} := \bigoplus_{i \in \mathcal{I}} \mathbb{C}^{\mathcal{T}_i}, \quad \mathcal{Y} := \mathbb{C}^{\mathcal{I}}. \quad (43)$$

Both spaces carry their raw counting-measure inner products. Attach a possibly complex unit gauge

$$\boldsymbol{\mu}_i = (\mu_{i,t})_{t \in \mathcal{T}_i}, \quad |\mu_{i,t}| = 1, \quad (44)$$

and define the erasure operator

$$(C_\mu z)_i := \sum_{t \in \mathcal{T}_i} \mu_{i,t} z_{i,t}. \quad (45)$$

The classical squarefree Möbius choice is real, but allowing a unitary gauge makes the conjugation in the spectral projector visible.

For each fiber define

$$\Pi_i z_i := \frac{\overline{\boldsymbol{\mu}_i}}{k_i} \sum_{t \in \mathcal{T}_i} \mu_{i,t} z_{i,t}. \quad (46)$$

Thus Π_i is the orthogonal projector onto $\text{span}\{\overline{\boldsymbol{\mu}_i}\}$, not in general onto $\text{span}\{\boldsymbol{\mu}_i\}$.

Theorem 3.1 (Exact erasure spectrum). *Let $C_{\mu,i} : \mathbb{C}^{\mathcal{T}_i} \rightarrow \mathbb{C}$ denote the i -th block of C_μ . Then*

$$C_{\mu,i}^* C_{\mu,i} = k_i \Pi_i, \quad (47)$$

$$C_\mu^* C_\mu = \bigoplus_{i \in \mathcal{I}} k_i \Pi_i, \quad (48)$$

$$\text{Ker } C_\mu = \bigoplus_{i \in \mathcal{I}} \left\{ z_i : \sum_{t \in \mathcal{T}_i} \mu_{i,t} z_{i,t} = 0 \right\}. \quad (49)$$

In particular,

$$\text{rank } C_\mu = |\mathcal{I}|, \quad \dim \text{Ker } C_\mu = \sum_{i \in \mathcal{I}} (k_i - 1). \quad (50)$$

The nonzero singular value on the i -th fiber is $\sqrt{k_i}$.

Proof. In the standard coordinate convention the row of $C_{\mu,i}$ is $\mu_i^\top$, so

$$C_{\mu,i}^* C_{\mu,i} = \overline{\mu_i} \mu_i^\top. \quad (51)$$

Its range is the line spanned by $\overline{\mu_i}$, and $\|\mu_i\|^2 = k_i$ by (44). This proves (47). Distinct fibers have disjoint domain and range coordinates, hence the direct-sum identity (48). Each nonzero row has rank one, and its kernel has dimension $k_i - 1$, which proves the remaining claims. $\square$

Remark 3.2 (The hypotheses behind the scalar k_i). For a nonzero general row $g_i = (g_{i,t})_t$, the exact identity is

$$C_i^* C_i = \overline{g_i} g_i^\top = \|g_i\|^2 \Pi_{\text{span}(\overline{g_i})}. \quad (52)$$

Thus the eigenvalue is k_i only when all k_i gauges have unit modulus. In particular, nonsquarefree labels carrying Möbius value zero cannot be counted among the k_i active unit-gauge labels. Also, when the numbers k_i vary, there is no global scalar k for which $C_\mu^* C_\mu = k \Pi$.

Proposition 3.3 (Sharp ambient obstruction). *The best lower erasure constant on the entire canonical space is*

$$\alpha_{\text{amb}} := \inf_{0 \neq z \in \mathcal{Z}} \frac{\|C_\mu z\|^2}{\|z\|^2} = \begin{cases} 1, & k_i = 1 \text{ for every } i, \\ 0, & k_i \geq 2 \text{ for at least one } i. \end{cases} \quad (53)$$

Both alternatives are sharp.

Proof. If every $k_i = 1$, then C_μ is diagonal with unit-modulus entries and is therefore unitary. If $k_i \geq 2$, choose distinct labels $t, u \in \mathcal{T}_i$, set

$$z_{i,t} = \mu_{i,u}, \quad z_{i,u} = -\mu_{i,t}, \quad (54)$$

and set every other coordinate to zero. This is a nonzero vector in $\text{Ker } C_\mu$, so the infimum vanishes. $\square$

The complex conjugation in (46) is not cosmetic. For example, if

$$C = \begin{bmatrix} 1 & i \end{bmatrix}, \quad (55)$$

then $\text{Ran } C^* = \text{span}\{(1, -i)\}$. Projecting instead onto $\text{span}\{(1, i)\}$ reverses both off-diagonal signs and does not give $C^* C$. Likewise, the row $\begin{bmatrix} 1 & 0 \end{bmatrix}$ has two displayed coordinates but nonzero eigenvalue one, not two; this is the smallest counterexample to omitting the unit-gauge condition.

3.2 Canonical aggregation is not physical uniqueness

We record the exact sense in which the residual coordinate system is canonical. For each i , let

$$\mathcal{T}_i = \mathcal{T}_i^0 \cup \mathcal{S}_i \quad (56)$$

be a declared disjoint partition into no-high labels and residual labels. For $t \in \mathcal{T}_i^0$ let $b_{i,t} \in \mathbb{C}$, and for $s \in \mathcal{S}_i$ let $\mathcal{P}_{i,s}$ be a finite set with coefficients $b_{i,s,p}$. Put

$$B_{i,s} := \sum_{p \in \mathcal{P}_{i,s}} b_{i,s,p}, \quad z_{i,t} := b_{i,t}, \quad z_{i,s} := -B_{i,s}. \quad (57)$$

Define

$$\mathcal{D}_0 := \sum_i \left(\sum_{t \in \mathcal{T}_i^0} |b_{i,t}|^2 + \sum_{s \in \mathcal{S}_i} \sum_{p \in \mathcal{P}_{i,s}} |b_{i,s,p}|^2 \right), \quad (58)$$

$$\mathcal{D}_H := \sum_i \sum_{s \in \mathcal{S}_i} \sum_{p \in \mathcal{P}_{i,s}} |b_{i,s,p}|^2, \quad (59)$$

$$\mathcal{G}_{\text{res}} := \sum_i \sum_{s \in \mathcal{S}_i} |B_{i,s}|^2. \quad (60)$$

Proposition 3.4 (Canonical residual identity). *For the coefficient vector in (57),*

$$\boxed{\mathcal{R}_{\text{low}} := \|z\|^2 = \mathcal{D}_0 - \mathcal{D}_H + \mathcal{G}_{\text{res}}.} \quad (61)$$

Equivalently,

$$\mathcal{R}_{\text{low}} = \mathcal{D}_0 + 2 \operatorname{Re} \sum_i \sum_{s \in \mathcal{S}_i} \sum_{\substack{p, q \in \mathcal{P}_{i,s} \\ p < q}} b_{i,s,p} \overline{b_{i,s,q}}, \quad (62)$$

after fixing any order on each finite prime set.

Proof. The declared partition (56) is orthogonal in $\mathcal{Z}$. Hence

$$\|z\|^2 = \sum_i \sum_{t \in \mathcal{T}_i^0} |b_{i,t}|^2 + \sum_i \sum_{s \in \mathcal{S}_i} |B_{i,s}|^2,$$

which is exactly (61). Expanding every $|B_{i,s}|^2$ and cancelling its diagonal contribution against $-\mathcal{D}_H$ gives (62). $\square$

The vector z is unique as the coefficient vector in the fixed orthonormal coordinate family (i, t) . This statement requires the partition in (56) to have been fixed first; it neither recovers the separate coefficients inside a sum $B_{i,s}$ nor makes z unique from $C_\mu z$. Its support is the smallest subset of this particular orthonormal coordinate family that represents z , but the fixed baseline label bank need not be minimal for one realized coefficient vector. Indeed,

$$b_{i,s,p} = 1, \quad b_{i,s,q} = -1 \quad (63)$$

gives $z_{i,s} = 0$, even though two baseline atoms occur. At the physical erasure level, the one-fiber vectors $(1, 0)$ and $(0, 1)$ have the same image under $\begin{bmatrix} 1 & 1 \end{bmatrix}$. Thus “canonical”, “minimal”, and “physically identifiable” are three different assertions.

3.3 The signed native off-diagonal

Put

$$(v_0)_i := \sum_{t \in \mathcal{T}_i^0} \mu_{i,t} b_{i,t}, \quad w_i := \sum_{s \in \mathcal{S}_i} \mu_{i,s} B_{i,s}. \quad (64)$$

Then the erasure of the canonical residual vector is

$$\boxed{C_\mu z = v_0 - w.} \quad (65)$$

The exact coordinate expansion is

$$\begin{aligned} \|C_\mu z\|^2 &= \|z\|^2 + 2 \operatorname{Re} \sum_i \sum_{\substack{t, u \in \mathcal{T}_i^0 \\ t < u}} \mu_{i,t} \overline{\mu_{i,u}} b_{i,t} \overline{b_{i,u}} \\ &\quad + 2 \operatorname{Re} \sum_i \sum_{\substack{s, r \in \mathcal{S}_i \\ s < r}} \mu_{i,s} \overline{\mu_{i,r}} B_{i,s} \overline{B_{i,r}} \\ &\quad - 2 \operatorname{Re} \sum_i \sum_{\substack{t \in \mathcal{T}_i^0 \\ s \in \mathcal{S}_i}} \mu_{i,t} \overline{\mu_{i,s}} b_{i,t} \overline{B_{i,s}}. \end{aligned} \quad (66)$$

Thus erasure restores all unequal-low-label cross terms. The mixed no-high/residual terms have a minus sign, while the no-high/no-high and residual/residual terms have a plus sign. For a complex gauge the factor is $\mu_{i,t}\overline{\mu_{i,u}}$; it cannot be replaced by a formal $\mu(tu)$ without additional coprimality and reality hypotheses.

If

$$(v_p)_i := \sum_{\substack{s \in \mathcal{S}_i \\ p \in \mathcal{P}_{i,s}}} \mu_{i,s} b_{i,s,p}, \quad (67)$$

then $w = \sum_p v_p$ and

$$\|v_0 - w\|^2 = \|v_0\|^2 + \sum_p \|v_p\|^2 + 2 \operatorname{Re} \sum_{p < q} \langle v_p, v_q \rangle - 2 \operatorname{Re} \sum_p \langle v_0, v_p \rangle. \quad (68)$$

No positive lower comparison with $\|z\|^2$ follows from the diagonal terms alone. The minimal sharp example is

$$C = \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad z = (1, -1), \quad \|z\|^2 = 2, \quad \|Cz\|^2 = 0. \quad (69)$$

The restricted-range analysis in the next section identifies exactly what additional coefficient geometry is needed to exclude this dark direction.

4 Restricted-range transfer and the three quantifiers

The ambient erasure constant vanishes as soon as one fiber contains two labels. A physical coefficient family may nevertheless occupy a smaller subspace transverse to the dark kernel. This section gives the exact finite-dimensional criterion, its Moore–Penrose form, and the distinction between an ambient theorem, a canonical-range theorem, and a statement for one distinguished vector.

4.1 Compression to the actual attainable range

Let $\mathcal{A}$ be a finite-dimensional Hilbert space, let

$$T : \mathcal{A} \longrightarrow \mathcal{Z}, \quad E := \operatorname{Ran} T, \quad (70)$$

and assume $E \neq \{0\}$. The map T is part of the data: replacing it by a free ambient coefficient array changes the quantifier and may create dark directions that the declared family never attains. Define

$$\alpha_E := \inf_{0 \neq z \in E} \frac{\|C_\mu z\|^2}{\|z\|^2}. \quad (71)$$

Theorem 4.1 (Exact restricted-range criterion). *Let P_E be the orthogonal projector onto E . Then*

$$\alpha_E = \lambda_{\min} \left((P_E C_\mu^* C_\mu P_E)|_E \right). \quad (72)$$

Here $\lambda_{\min}$ is the ordinary smallest eigenvalue, including zero. Moreover,

$$\alpha_E > 0 \iff E \cap \operatorname{Ker} C_\mu = \{0\}. \quad (73)$$

Proof. For $z \in E$,

$$\langle z, P_E C_\mu^* C_\mu P_E z \rangle = \langle z, C_\mu^* C_\mu z \rangle = \|C_\mu z\|^2. \quad (74)$$

The compression in (72) is a positive self-adjoint operator on the finite-dimensional space E . The Rayleigh–Ritz principle proves the eigenvalue formula. Its smallest eigenvalue is zero exactly when some nonzero $z \in E$ has $C_\mu z = 0$, which is (73). $\square$

The finite-dimensional hypothesis is essential for the reverse implication in (73). On $\ell^2(\mathbb{N})$, the injective diagonal operator

$$C(x_1, x_2, \dots) = (x_1, x_2/2, x_3/3, \dots) \quad (75)$$

has trivial kernel but is not bounded below. Even in finite dimension one must use the smallest eigenvalue rather than the smallest *positive* eigenvalue: for

$$C_\mu = \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad E = \mathbb{C}^2, \quad (76)$$

the smallest positive eigenvalue of $C_\mu^* C_\mu$ is two, while $\alpha_E = 0$.

4.2 The support-restricted pseudoinverse formula

Set

$$R := T^*T, \quad N := T^*C_\mu^*C_\mu T, \quad S := \text{Ran } R = (\text{Ker } T)^\perp. \quad (77)$$

Let $R^\dagger$ be the Moore–Penrose pseudoinverse and write $R^{\dagger/2} := (R^\dagger)^{1/2}$.

Theorem 4.2 (Range-normalized Gram formula). *The operator*

$$U := TR^{\dagger/2}|_S : S \longrightarrow E \quad (78)$$

is unitary. Consequently,

$$\boxed{\alpha_E = \lambda_{\min}((R^{\dagger/2}NR^{\dagger/2})|_S)}. \quad (79)$$

If

$$\Theta := R^{\dagger/2}(N - R)R^{\dagger/2}, \quad (80)$$

then the equivalent defect formula is

$$\boxed{\alpha_E = 1 + \lambda_{\min}(\Theta|_S)}. \quad (81)$$

The restriction to S in both formulas is mandatory.

Proof. On S , functional calculus for the positive operator R gives

$$U^*U = R^{\dagger/2}T^*TR^{\dagger/2}|_S = R^{\dagger/2}RR^{\dagger/2}|_S = I_S. \quad (82)$$

The range of U equals $\text{Ran } T = E$, either by the polar decomposition or directly on each positive eigenspace of R . Hence U is unitary from S onto E . Furthermore,

$$U^*C_\mu^*C_\mu U = (R^{\dagger/2}NR^{\dagger/2})|_S. \quad (83)$$

The right side is therefore unitarily equivalent to the compressed operator in (72), proving (79). Finally,

$$R^{\dagger/2}RR^{\dagger/2}|_S = I_S, \quad (84)$$

so subtracting this identity gives (81). $\square$

For reference, the same calculation gives the standard exact formulas

$$T^\dagger = R^\dagger T^*, \quad P_E = TT^\dagger = TR^\dagger T^*. \quad (85)$$

Taking the minimum in (81) on the full parameter space $\mathcal{A}$ is generally false, because Θ vanishes on $\text{Ker } R$. The smallest counterexample is

$$T(a, b) = \frac{a}{\sqrt{2}}(1, 1), \quad C_\mu = \begin{bmatrix} 1 & 1 \end{bmatrix}. \quad (86)$$

Here

$$R = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad N = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}, \quad \Theta = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \quad (87)$$

The true restricted constant is two, whereas $1 + \lambda_{\min}(\Theta) = 1$ on all of $\mathbb{C}^2$.

4.3 Dimension kill and its nonconverse

Proposition 4.3 (One-way dimension kill). *If*

$$\dim E > \text{rank } C_\mu, \quad (88)$$

then $\alpha_E = 0$. The converse is false.

Proof. The restriction $C_\mu|_E : E \rightarrow \mathcal{Y}$ has rank at most $\text{rank } C_\mu$. Under (88), rank-nullity supplies a nonzero element of $E \cap \text{Ker } C_\mu$, and [theorem 4.1](#) gives $\alpha_E = 0$.

For the nonconverse, take $C_\mu = \begin{bmatrix} 1 & 1 \end{bmatrix}$ and $E = \text{span}\{(1, -1)\}$. Then $\dim E = \text{rank } C_\mu = 1$, but $E \subseteq \text{Ker } C_\mu$ and $\alpha_E = 0$. $\square$

In the erasure geometry of [theorem 3.1](#), $\text{rank } C_\mu = |\mathcal{I}|$. If the raw grouping map is surjective onto the full canonical space, then $E = \mathcal{Z}$, and any fiber with $k_i > 1$ activates the dark space. Thus a positive restricted theorem must use a genuinely smaller *actual attainable range*; it cannot be obtained by declaring all canonical coordinates to be free and then forgetting that enlargement.

4.4 Ambient, canonical-range, and distinguished statements

There are three different quantifiers:

$$\alpha_{\text{amb}} := \inf_{0 \neq z \in \mathcal{Z}} \frac{\|C_\mu z\|^2}{\|z\|^2}, \quad (89)$$

$$\alpha_E := \inf_{0 \neq z \in E} \frac{\|C_\mu z\|^2}{\|z\|^2}, \quad (90)$$

$$\rho(z_*) := \frac{\|C_\mu z_*\|^2}{\|z_*\|^2} \quad (0 \neq z_* \in E). \quad (91)$$

They obey only the one-way chain

$$\boxed{\alpha_{\text{amb}} \leq \alpha_E \leq \rho(z_*)}. \quad (92)$$

The two reverse implications fail sharply. With $C = \begin{bmatrix} 1 & 1 \end{bmatrix}$, the ambient constant is zero, while for

$$E = \text{span}\{(1, 1)\} \quad (93)$$

one has $\alpha_E = 2$. On the other hand, with the same C and $E = \mathbb{C}^2$, one has $\alpha_E = 0$, while the distinguished vector $z_* = (1, 1)$ satisfies $\rho(z_*) = 2$. Consequently:

- (Q1) an ambient lower bound controls every canonical family and every distinguished vector;
- (Q2) a canonical-range lower bound controls only vectors proved to lie in that one declared range;
- (Q3) a positive distinguished ratio does not imply a uniform theorem on its containing range.

Calling one of these statements by the name of another is a quantifier change, not a harmless normalization.

4.5 The independent affine gate

The restricted-range constant resolves only the erasure from a residual coefficient vector to its native high component. It does not prevent that component from cancelling a no-high vector. Abstractly, let $\mathcal{Z}_H \subseteq \mathcal{Z}$, let $E_H = \text{Ran } T_H \subseteq \mathcal{Z}_H$ be nonzero, and define

$$\alpha_H := \inf_{0 \neq r \in E_H} \frac{\|C_\mu r\|^2}{\|r\|^2}. \quad (94)$$

For one signed residual vector $r \in E_H$, put

$$h := C_\mu r, \quad \mathcal{G}_{\text{res}} := \|r\|^2, \quad \mathcal{N}_H := \|h\|^2. \quad (95)$$

Proposition 4.4 (Two independent transfer gates). *Let $v_0 \in \mathcal{Y}$. Suppose $r \neq 0$ and, independently, that*

$$\|v_0 + h\|^2 \geq \delta_{\text{aff}}(\|v_0\|^2 + \|h\|^2) \quad (\delta_{\text{aff}} \geq 0). \quad (96)$$

Then

$$\boxed{\|v_0 + h\|^2 \geq \delta_{\text{aff}} \mathcal{N}_H \geq \delta_{\text{aff}} \alpha_H \mathcal{G}_{\text{res}}.} \quad (97)$$

Neither inequality supplies the hypothesis of the other.

Proof. The first inequality follows from $\|v_0\|^2 + \|h\|^2 \geq \|h\|^2 = \mathcal{N}_H$. The definition of α_H gives $\mathcal{N}_H \geq \alpha_H \mathcal{G}_{\text{res}}$, proving the chain.

For independence, take the one-dimensional example

$$r = 1, \quad C_\mu r = h = 1, \quad v_0 = -1. \quad (98)$$

The restricted erasure constant and both high energies equal one, but $\|v_0 + h\|^2 = 0$, so no positive affine constant follows. Conversely, an affine separation for one chosen pair says nothing about the infimum of C_μ on all of E_H . $\square$

For a class $\mathcal{F} \subseteq \mathcal{Y} \oplus \mathcal{Y}$ of admissible pairs, the exact affine constant is

$$\delta_{\text{aff}} = \inf_{\substack{(v_0, h) \in \mathcal{F} \\ \|v_0\|^2 + \|h\|^2 > 0}} \frac{\|v_0 + h\|^2}{\|v_0\|^2 + \|h\|^2}. \quad (99)$$

It lies in $[0, 2]$, and a positive value is precisely a uniform exclusion of approach to the cancellation direction $v_0 = -h$. For an arbitrary, possibly nonclosed class $\mathcal{F}$, absence of one exact anti-diagonal pair does not suffice: normalized admissible pairs may still converge to the anti-diagonal. If $\mathcal{F}$ is a nonzero finite-dimensional linear subspace, as is the actual coupled image used below, compactness of its unit sphere makes positivity equivalent to trivial intersection with the anti-diagonal. This is a separate signed off-diagonal problem; the spectrum of the residual erasure operator alone does not solve it.

5 The literal fixed-shift residual-to-native interface

We now identify the high residual vector in [section 2](#) with the literal terminal output of TPC-53-57. This is a coordinatewise interface theorem at the one specified h_0 . It is not a lower Gram estimate.

5.1 One side, one source system, and one physical intercept

Fix one side of the equality column. On the left, a terminal atom is

$$\mathbf{u} = (m, r, j, \gamma), \quad m = d_m \ell_m, \quad p_u r - m j = h_0, \quad (100)$$

and its native and residual labels are

$$i(\mathbf{u}) := (L, \gamma, j), \quad s(\mathbf{u}) := d_m r. \quad (101)$$

The right side has $\mathbf{u} = (m, r, j, \alpha)$, with the source systems interchanged, and $i(\mathbf{u}) = (R, \alpha, j)$. Every assertion below is first proved on one side. The two-sided assertion is the orthogonal direct sum of the two one-sided maps; it is not a new coherent sum.

The row, mask, and scale data are the literal ones from TPC-53:

$$m = d_m \ell_m, \quad a_m = \mu(d_m)(\log \ell_m) \omega_D(d_m) \psi_L(\ell_m/L) \zeta_m, \quad (102)$$

$$K_{m\gamma} = \mathbf{1}_{\ell_m \neq \ell_\gamma} \mathbf{1}_{|m-m_\gamma| > \Delta} \mathbf{1}_{(d_m, d_\gamma) \leq G}, \quad \Delta = QX^{-1/400}, \quad G = X^{1/400}. \quad (103)$$

The fixed opposite-row factor, orbit scalar, and four-factor profile are also retained. We abbreviate their exact product by

$$\mathfrak{P}_{m,\gamma}(j) := \eta_\gamma \mathcal{R}(q_\gamma) \xi_a W_1(mj/X) \Psi_1(d_m j/K) W_2(m_\gamma j/X) \Psi_2(d_\gamma j/K). \quad (104)$$

Thus the pre-terminal coordinate is

$$[G_X^L]_{\mathbf{u}} = a_m K_{m\gamma} \mathfrak{P}_{m,\gamma}(j). \quad (105)$$

Use the normalized shaped terminal weight

$$\Lambda_X(n) := (\log n) \Phi_0(n/X). \quad (106)$$

This is the same inherited cutoff as the TPC-45 notation $\Lambda(n) = (\log n) \Phi_0(n)$ after its scale has been absorbed into the profile. We use (106) throughout and do not identify it with the von Mangoldt function.

Let $\mathcal{U}_{h_0}^{\text{term}}$ be the exact terminal-admissible set: the candidate $p_u = (mj + h_0)/r$ is prime in the inherited support, $\Lambda_X(p_u r) \neq 0$, and

$$d_m, r, s = d_m r \text{ are squarefree}, \quad (d_m, r) = 1. \quad (107)$$

For comparison with the TPC-45 very-high Gram, make the deterministic restriction

$$\mathcal{U}_{h_0, > y}^{\text{term}} := \{\mathbf{u} \in \mathcal{U}_{h_0}^{\text{term}} : p_u > y\}, \quad (108)$$

where y obeys (17)–(18). The conclusions in this section concern this very-high face. They do not silently cover the remainder of the large-prime band. At the endpoint cutoff, the cofactor ceiling (20) also makes this face disjoint from the old $R \asymp J$ block. Nontrivial activation on a compatible smaller cofactor band is an additional arithmetic hypothesis, not a consequence of the coordinate dictionary.

5.2 Baseline and distinguished supports

The exact factorization of TPC-56–57 has the form

$$X_{\mathbf{u}} := (D_{h_0} G_X^L)_{\mathbf{u}} = c_{\mathbf{u}} \zeta_{m(\mathbf{u})}, \quad (109)$$

where c_u contains the common output factor, literal mask, all displayed profiles, source shapes, and $\Lambda_X(mj + h_0)$, but not the bounded row coefficient ζ_m . Define the baseline and physical very-high carriers separately:

$$\mathcal{U}_{h_0, > y}^o := \{u \in \mathcal{U}_{h_0, > y}^{\text{term}} : c_u \neq 0\}, \quad (110)$$

$$\mathcal{U}_{h_0, > y}^{\text{phys}}(\zeta) := \{u \in \mathcal{U}_{h_0, > y}^o : \zeta_{m(u)} \neq 0\}. \quad (111)$$

The Hilbert spaces and the erasure map are built on the fixed baseline carrier (110). A distinguished coefficient vector is represented by zeros on inactive baseline coordinates. This keeps coefficient-dependent cancellation separate from structural deletion.

TPC-55 proves, under $D = o(J)$, row-factorization uniqueness, and large X , that

$$u \mapsto (i(u), s(u), p_u) \quad (112)$$

is injective on one declared source system. Thus a coordinate (i, s, p) denotes at most one literal atom. This is the canonical nonduplicated prime-resolved representation on the fixed baseline carrier used below; it is not a claim of minimal parameter dimension or physical identifiability [4].

5.3 Sign calibration against the stripped TPC-45 atom

Let $b_{i, sp}$ be the stripped TPC-45 coefficient on the coordinate corresponding through (112). The exact factor allocation of TPC-53 puts $\mu(d_m)$ in the source coefficient and the terminal operator contributes $\mu(r)$. Hence

$$\mu(d_m)\mu(r) = \mu(s) \quad (113)$$

on (107). With the leading sign in the TPC-45 stripped atom fixed as in that paper, coordinate-wise comparison of the source coefficient, mask, opposite-row factor, orbit scalar, profiles, and shaped terminal weight gives

$$\boxed{X_u = -\mu(s(u))b_{i(u), s(u)p_u}.} \quad (114)$$

No modulus has been taken in (114). It is the coefficientwise global-sign calibration needed for the later base-high affine cross term [2, 3].

For clarity, define the stripped grouped high vector directly from the TPC-45 atoms by

$$z_H(i, s) := - \sum_{\substack{p > y \\ sp \in \mathcal{U}_i^o}} b_{i, sp}. \quad (115)$$

This is exactly the convention $z_H = -\sum_p x_p$ of (33). Put

$$A_H := U_\mu z_H, \quad A_H(i, s) = \mu(s)z_H(i, s). \quad (116)$$

Then (114) and the prime-resolved injection show

$$\boxed{A_H(i, s) = \sum_{\substack{u \in \mathcal{U}_{h_0, > y}^{\text{term}} \\ i(u)=i, s(u)=s}} X_u.} \quad (117)$$

Thus A_H , not the stripped z_H , is the TPC-57 terminal grouped amplitude. In particular, the TPC-57 amplitude already contains the common Möbius factor $\mu(s)$; applying U_μ to it a second time would remove rather than restore the physical residual sign.

Proposition 5.1 (Literal fixed-shift erasure interface). *On one declared side and source system, under the cutoff conditions (17)–(18) and the TPC–55 injection, the following diagram commutes:*

$$\begin{aligned} \{\text{prime-resolved atoms } (i, s, p)\} &\xrightarrow{p\text{-grouping}} A_H \in \ell^2(i, s) \\ &\downarrow \mathbf{E} \\ &= \sum_{p \in \mathcal{P}_y} v_p \in \ell^2(i). \end{aligned} \quad (118)$$

Equivalently,

$$\boxed{\mathbf{E}A_H = - \sum_{p \in \mathcal{P}_y} v_p.} \quad (119)$$

Every arrow is pointwise at the one physical h_0 ; no average over intercepts and no frozen-shift completion occurs.

Proof. The first arrow is (117). Applying the unsigned erasure and using (114) gives, at a native coordinate i ,

$$(\mathbf{E}A_H)_i = - \sum_{p > y} \sum_s \mu(s) b_{i,sp} = - \sum_{p > y} (v_p)_i,$$

which is (119). All sums are finite and retain the same literal carrier. $\square$

5.4 Raw norms and the second grouping gate

The terminal atomic, residual grouped, and native-high energies on this face are three different raw counting-measure quantities:

$$\mathcal{D}_{h_0}^{>y}(G_X^L) := \sum_{\mathbf{u} \in \mathcal{U}_{h_0, >y}^{\text{term}}} |X_{\mathbf{u}}|^2, \quad (120)$$

$$\mathcal{G}_{h_0}^{>y}(G_X^L) := \sum_{i,s} \left| \sum_{\mathbf{u}: i(\mathbf{u})=i, s(\mathbf{u})=s} X_{\mathbf{u}} \right|^2 = \|A_H\|_{\mathcal{H}_{\text{res}}^H}^2, \quad (121)$$

$$\mathcal{N}_H^{>y}(h_0) := \sum_i \left| \sum_s A_H(i, s) \right|^2 = \|\mathbf{E}A_H\|_{\mathcal{H}_{\text{nat}}}^2. \quad (122)$$

Equation (121) already forms the coherent terminal-prime sum at fixed (i, s) . The input to the native Gram gate is therefore $\mathcal{G}_{h_0}^{>y}$, not $\mathcal{D}_{h_0}^{>y}$. The passage from (121) to (122) is the second grouping $(i, s) \rightarrow i$.

The singleton and phase-sector results of TPC–56–57 concern the first grouping $p \rightarrow (i, s)$ [5, 6]. They do not constrain the cross- s sum in (122). In particular:

- (i) a terminal-prime singleton at (i, s) may coexist with many other singleton residual labels $t \neq s$ in the same native coordinate;
- (ii) the adaptive phase sector of TPC–56 deletes terminal coordinates before the first grouping and is not the undeleted vector A_H ;
- (iii) the conditional undeleted floor of TPC–57 can provide a lower bound for $\|A_H\|^2$, but no lower bound for $\|\mathbf{E}A_H\|^2$ follows without a new residual-label comparison.

Finally, (122) is only the very-high part of the native affine column on the declared direct slice. Its complete native affine energy contains

$$v_0 + \mathbf{E}A_H = v_0 - \sum_{p \in \mathcal{P}_y} v_p. \quad (123)$$

Neither nonnegativity nor a triangle inequality permits a lower bound for $\|\mathbf{E}A_H\|^2$ to be returned to $\|v_0 + \mathbf{E}A_H\|^2$. Base completion is a separate affine gate.

The fixed-shift interface proved here is therefore exact but deliberately limited: it aligns literal coefficients, signs, supports, and raw coordinates through both grouping maps. It proves no positive lower singular value for $\mathbf{E}$, no terminal activation, no residual-label occupancy estimate, and no native affine lower bound.

6 High-block transfer and completion of the affine face

The preceding residual construction and the native affine column use two equivalent Möbius conventions. We first align them exactly. This is essential: the terminal amplitudes of TPC–56 and TPC–57 already contain the residual sign, whereas the canonical Walsh lift stores that sign in the evaluation operator [2, 5, 6].

6.1 The sign-compatible high block

Let i denote one native output coordinate, including its side, opposite row, and orbit time. For a squarefree low label s , define the canonical high Walsh coefficient

$$z_{H;i,s} := - \sum_{\substack{p \in \mathcal{P}_y \\ sp \text{ is an active equality label at } i}} b_{i,sp}. \quad (124)$$

The minus sign is the fixed physical value of the high character. Let

$$(C_{\mu,H}z)_i := \sum_s \mu(s) z_{i,s}, \quad (U_\mu z)_{i,s} := \mu(s) z_{i,s}. \quad (125)$$

Both operators act on raw counting-measure spaces. Put

$$h := C_{\mu,H} z_H. \quad (126)$$

By the definition of the TPC–45 vectors v_p ,

$$\boxed{h = - \sum_{p \in \mathcal{P}_y} v_p.} \quad (127)$$

For the same very-high-prime carrier, let

$$A_{i,s} := \sum_{u:q(u)=(i,s)} x_u \quad (128)$$

be the undeleted TPC–57 residual amplitude. Here the side tag in i is suppressed from the notation $q(u)$. Finally define the unsigned erasure

$$(C_1 A)_i := \sum_s A_{i,s}. \quad (129)$$

Proposition 6.1 (Exact gauge alignment). *On the identical fixed- h_0 , very-high-prime carrier,*

$$\boxed{A = U_\mu z_H, \quad C_1 A = C_{\mu,H} z_H = h, \quad \|A\|_2 = \|z_H\|_2.} \quad (130)$$

Equivalently, $C_{\mu,H} = C_1 U_\mu$. In particular, no second Möbius sign is to be applied to the already physical amplitudes A .

Proof. Write an active high equality label as sp , with $s = d_m r$. The stripped TPC-45 atom has the displayed leading minus sign, while the terminal multiplier in TPC-56 and TPC-57 contributes $\mu(d_m)\mu(r) = \mu(s)$. Thus, after summing all terminal primes with the same (i, s) ,

$$A_{i,s} = -\mu(s) \sum_p b_{i,sp} = \mu(s) z_{H;i,s}.$$

This proves the first identity. Summing over s proves the second, and the third follows because U_μ is diagonal unitary. $\square$

Let z_0 be the no-high part of the canonical Walsh lift, so that $(z_0)_{i,u} = b_{i,u}$ when $u_{>y} = 1$. Then

$$C_{\mu,0}z_0 = v_0, \quad C_{\mu,H}z_H = h, \quad C_\mu(z_0 \oplus z_H) = v_0 + h. \quad (131)$$

Consequently the complete native affine energy is

$$\boxed{\mathcal{N}_{\text{aff}} := \|v_0 + h\|_2^2 = \left\| v_0 - \sum_p v_p \right\|_2^2.} \quad (132)$$

6.2 The restricted high-transfer constant

For coefficient-uniform statements, fix once and for all a baseline carrier independent of the bounded row coefficient vector $\zeta = (\zeta_m)$. Remove atoms whose baseline factor is identically zero, but do not remove an atom merely because a particular ζ_m vanishes. On this fixed carrier, let

$$T_H : \mathcal{K}_X \longrightarrow \mathcal{H}_H, \quad T_H \zeta = z_H(\zeta), \quad E_H := \text{Ran } T_H, \quad (133)$$

where $\mathcal{K}_X$ is the finite row-coefficient space and $\mathcal{H}_H$ is the canonical high-label Hilbert space. The exact literal factorization $x_u = c_u \zeta_{m(u)}$ makes T_H linear. For the coefficient-uniform gate, assume $E_H \neq \{0\}$ and define

$$\alpha_H := \inf_{\substack{z \in E_H \\ z \neq 0}} \frac{\|C_{\mu,H}z\|_2^2}{\|z\|_2^2}. \quad (134)$$

If $E_H = \{0\}$, the declared high carrier is inactive for the whole coefficient class. The pointwise identities above still hold, but no least-eigenvalue constant α_H is assigned and the two-stage lower chain has no nonzero high input.

Proposition 6.2 (Optimal high-block transfer). *Let P_{E_H} be the orthogonal projection onto E_H . Then*

$$\boxed{\alpha_H = \lambda_{\min}(P_{E_H} C_{\mu,H}^* C_{\mu,H} P_{E_H}|_{E_H}).} \quad (135)$$

Moreover,

$$\alpha_H > 0 \iff E_H \cap \text{Ker } C_{\mu,H} = \{0\}. \quad (136)$$

In particular,

$$\dim E_H > \text{rank } C_{\mu,H} \implies \alpha_H = 0. \quad (137)$$

The same value is obtained if the infimum is restricted to any bounded coefficient cube containing a nonzero scalar multiple of every direction in $\mathcal{K}_X$.

Proof. The quadratic form of the compressed positive operator in (135) is exactly $\|C_{\mu,H}z\|_2^2$ on E_H . The Rayleigh–Ritz principle therefore gives the first assertion. In finite dimension its least eigenvalue is zero exactly when the restriction of $C_{\mu,H}$ to E_H has a nonzero kernel. Rank–nullity proves (137). Finally, every nonzero coefficient vector can be multiplied by a sufficiently small positive scalar to enter the bounded cube, and the quotient (134) is homogeneous of degree zero. $\square$

For the distinguished coefficient vector alone, no uniform coefficient class is needed. Whenever $z_H \neq 0$, set

$$\alpha_H(\zeta) := \frac{\|h(\zeta)\|_2^2}{\|z_H(\zeta)\|_2^2}. \quad (138)$$

This number is an identity, not an estimated lower bound.

6.3 The affine-separation constant

Let

$$B_0 : \mathcal{K}_X \longrightarrow \mathcal{H}_{\text{out}}, \quad B_0 \zeta = v_0(\zeta), \quad (139)$$

denote the no-high native map on the same declared coefficient class, and define the coupled map

$$J_{\text{aff}} : \mathcal{K}_X \longrightarrow \mathcal{H}_{\text{out}} \oplus \mathcal{H}_{\text{out}}, \quad J_{\text{aff}} \zeta := (B_0 \zeta, C_{\mu, H} T_H \zeta). \quad (140)$$

Both entries are therefore driven by the same coefficient vector. Let

$$\mathcal{F}_{\text{aff}} := \text{Ran } J_{\text{aff}} \subset \mathcal{H}_{\text{out}} \oplus \mathcal{H}_{\text{out}}. \quad (141)$$

For the uniform affine gate assume $\mathcal{F}_{\text{aff}} \neq \{0\}$. If this image is zero, the declared native affine column vanishes identically; no restricted minimum δ_{aff} is assigned and no positive lower transfer is available on that class. Define the physical sum map

$$D_+(b, g) := b + g \quad (142)$$

and the optimal affine-separation constant

$$\delta_{\text{aff}} := \inf_{\substack{(b, g) \in \mathcal{F}_{\text{aff}} \\ (b, g) \neq (0, 0)}} \frac{\|b + g\|_2^2}{\|b\|_2^2 + \|g\|_2^2}. \quad (143)$$

Thus $0 \leq \delta_{\text{aff}} \leq 2$. Notice that this is the raw physical constant. Dividing D_+ by $\sqrt{2}$ would produce a normalized principal-angle constant, but it would also change the physical norm.

Proposition 6.3 (Exact affine kernel test). *One has*

$$\boxed{\delta_{\text{aff}} = \sigma_{\min}(D_+|_{\mathcal{F}_{\text{aff}}})^2}, \quad (144)$$

where the domain has the orthogonal-sum norm. Hence

$$\delta_{\text{aff}} > 0 \iff \mathcal{F}_{\text{aff}} \cap \{(g, -g) : g \in \mathcal{H}_{\text{out}}\} = \{(0, 0)\}. \quad (145)$$

The equivalence uses that $\mathcal{F}_{\text{aff}}$ is a nonzero finite-dimensional linear subspace, and hence closed; it is not asserted for an arbitrary nonclosed admissible set. In particular,

$$\dim \mathcal{F}_{\text{aff}} > \dim \mathcal{H}_{\text{out}} \implies \delta_{\text{aff}} = 0. \quad (146)$$

Proof. The quotient in (143) is the squared Rayleigh quotient of $D_+|_{\mathcal{F}_{\text{aff}}}$. This proves (144). Its kernel is the anti-diagonal $\{(g, -g)\}$, which proves the second assertion. The last follows from rank-nullity because D_+ maps into $\mathcal{H}_{\text{out}}$. $\square$

Theorem 6.4 (Two-stage residual-to-affine transfer). *Assume $E_H \neq \{0\}$ and $\mathcal{F}_{\text{aff}} \neq \{0\}$. For every coefficient vector in the declared baseline class,*

$$\boxed{\mathcal{N}_{\text{aff}}(\zeta) \geq \delta_{\text{aff}} \alpha_H \|z_H(\zeta)\|_2^2}. \quad (147)$$

Equivalently, using the TPC-57 residual vector $A = U_\mu z_H$,

$$\mathcal{N}_{\text{aff}}(\zeta) \geq \delta_{\text{aff}} \alpha_H \|A(\zeta)\|_2^2. \quad (148)$$

For one distinguished vector the same conclusion holds with the pointwise ratios $\alpha_H(\zeta)$ and

$$\delta_{\text{aff}}(\zeta) := \frac{\|v_0(\zeta) + h(\zeta)\|_2^2}{\|v_0(\zeta)\|_2^2 + \|h(\zeta)\|_2^2}, \quad (149)$$

whenever their denominators are nonzero.

Proof. By definition of δ_{aff} ,

$$\|v_0 + h\|_2^2 \geq \delta_{\text{aff}} (\|v_0\|_2^2 + \|h\|_2^2) \geq \delta_{\text{aff}} \|h\|_2^2.$$

By (134),

$$\|h\|_2^2 = \|C_{\mu,H} z_H\|_2^2 \geq \alpha_H \|z_H\|_2^2.$$

Multiplication proves (147), and proposition 6.1 gives (148). The distinguished statement is the same chain with the two defining identities inserted pointwise. $\square$

6.4 Two independent sharp obstructions

Theorem 6.5 (Sharp independence of the two transfer gates). *Neither stage in theorem 6.4 follows from positivity at the other stage.*

- (i) *A residual packet may have positive energy, with every terminal-prime fiber a singleton, while its native high vector is zero.*
- (ii) *A native high vector may be nonzero while the complete affine vector is zero.*

Both statements are sharp finite-dimensional assertions with raw counting measure.

Proof. Both examples use one common coefficient space $\mathcal{K}_X = \mathbb{C}$, so the two affine entries are genuinely coupled rather than selected independently.

For (i), take one native coordinate i , two distinct residual labels s_1, s_2 , and one terminal atom in each residual fiber. Let $z_d \neq 0$ be the dark vector

$$(z_d)_{i,s_1} = \overline{\mu(s_1)}, \quad (z_d)_{i,s_2} = -\overline{\mu(s_2)}, \quad C_{\mu,H} z_d = 0.$$

Define $T_H \zeta = \zeta z_d$ and, for a fixed nonzero $g \in \mathcal{H}_{\text{out}}$, define $B_0 \zeta = \zeta g$. Then $E_H = \text{span}\{z_d\}$, so $\alpha_H = 0$, whereas

$$\mathcal{F}_{\text{aff}} = \text{span}\{(g, 0)\}, \quad \delta_{\text{aff}} = 1.$$

Thus a strictly positive affine gate does not prevent total loss at native erasure, even with singleton terminal-prime fibers.

For (ii), choose a bright vector z_b with $C_{\mu,H} z_b = g \neq 0$, and set $T_H \zeta = \zeta z_b$, $B_0 \zeta = -\zeta g$. Then

$$\alpha_H = \frac{\|g\|_2^2}{\|z_b\|_2^2} > 0, \quad \mathcal{F}_{\text{aff}} = \text{span}\{(-g, g)\}, \quad \delta_{\text{aff}} = 0.$$

Hence a positive high-transfer gate does not prevent exact base-high cancellation. These are complete finite-dimensional coupled-map models; they do not assert that either bad direction occurs in the distinguished arithmetic image. $\square$

The last sentence is part of the theorem boundary. In particular, a dark direction on the unrestricted coordinate space is not evidence that the distinguished arithmetic coefficient occupies it. Conversely, an upper bound for fiber multiplicity does not bound its angle away from that dark space.

7 Transport boundaries, the signed native exchange, and the endpoint ledger

The exact transfer theorem does not make either restricted-range constant positive. This section explains why the two preceding papers do not supply that positivity, identifies the literal signed quadratic form that would supply it, and records the complete endpoint charge.

7.1 Why the adaptive phase sector does not transport

TPC-56 extracts a constant-energy subpacket after the terminal phases have been observed. Its selector is a genuine coordinate projection once it is chosen, but it is not a coefficient-independent pre-terminal criterion [5]. More basically, squared norms are not monotone under putting the deleted coordinates back into a coherent sum.

Proposition 7.1 (Sharp adaptive nontransport). *There is no constant $c > 0$ for which the grouped energy of the undeleted residual packet, or the complete native affine energy of the declared slice, is bounded below by c times the grouped energy of every phase-adaptively selected terminal subpacket. This failure already occurs in one two-atom residual fiber.*

Proof. Take two terminal coordinates in one residual fiber with amplitudes 1 and -1 . They lie in distinct phase sectors. A sector containing either coordinate has terminal diagonal and selected grouped energy equal to one, whereas restoring the other coordinate makes the undeleted grouped amplitude zero. After a common phase rotation avoiding sector boundaries, the two antipodal amplitudes occupy distinct members of the five equal sectors of TPC-56. Since the native and affine maps are downstream of this already vanishing undeleted residual amplitude, no positive lower bound for either downstream quantity can follow. The zero lower endpoint is attained, so the obstruction is sharp. $\square$

The proposition does not weaken the TPC-56 selected-square theorem. It states that the selected theorem may enter a physical chain only if the same selection is realized before terminal primality and is preserved by every later operation, or if a separate lower-reassembly theorem is proved. No such realization is assumed here.

7.2 What the singleton theorem does and does not preserve

TPC-57 instead works with the undeleted residual packet. For $y = (i, s)$, let

$$A_{i,s} := \sum_{u \in \mathcal{F}_y^{\text{term}}} x_u, \quad \mathcal{G}_{\text{res}} := \sum_{i,s} |A_{i,s}|^2. \quad (150)$$

Its singleton theorem applies to the first grouping

$$\{\text{terminal primes}\} \longrightarrow (i, s). \quad (151)$$

The native high column applies the different, second grouping

$$(i, s) \longrightarrow i, \quad H_i := \sum_s A_{i,s}. \quad (152)$$

Proposition 7.2 (Singleton survival stops at the residual label). *If a terminal-prime fiber $y = (i, s)$ is a physical singleton, then its energy survives the first grouping exactly. Even if every active first-stage fiber is a singleton, however, there is no positive support-only lower bound for*

$$\mathcal{N}_H := \sum_i |H_i|^2 \quad (153)$$

in terms of $\mathcal{G}_{\text{res}}$.

Proof. The first assertion is the identity $\left| \sum_{u \in \mathcal{F}_y} x_u \right|^2 = |x_u|^2$ on a singleton. For the second, take one native output i , two distinct residual labels s_1, s_2 , and one terminal atom in each residual fiber. Choose their already physical amplitudes to be $A_{i,s_1} = 1$ and $A_{i,s_2} = -1$. Then $\mathcal{G}_{\text{res}} = 2$, while $H_i = 0$ and $\mathcal{N}_H = 0$. Thus perfect singleton occupancy at the first grouping does not control the second grouping. $\square$

Accordingly, the singleton exponent of TPC-57 and the native high-transfer exponent below are separate entries. The factor-exchange theorem of TPC-57 studies collisions with equal s ; the new native gate consists of cross terms with unequal s .

7.3 The exact signed native exchange

Return to the literal fixed- h_0 amplitudes

$$x_u = c_u \zeta_{m(u)}, \quad q(u) = (i, s(u)), \quad s(u) = d_{m(u)} r_u, \quad (154)$$

where i contains the side, opposite row, and orbit time. Every active atom satisfies

$$p_u r_u - m(u) j_u = h_0. \quad (155)$$

Choose any fixed total order on the finite residual-label set in each native fiber.

Proposition 7.3 (Exact native exchange identity). *The difference between the native high energy and the residual grouped energy is*

$$\mathcal{N}_H - \mathcal{G}_{\text{res}} = 2 \operatorname{Re} \sum_i \sum_{s < t} A_{i,s} \overline{A_{i,t}} \quad (156)$$

$$= 2 \operatorname{Re} \sum_i \sum_{s < t} \sum_{u: q(u)=(i,s)} \sum_{v: q(v)=(i,t)} c_u \overline{c_v} \zeta_{m(u)} \overline{\zeta_{m(v)}}. \quad (157)$$

In every summand of (157), both atoms satisfy (155), but

$$d_{m(u)} r_u = s \neq t = d_{m(v)} r_v. \quad (158)$$

Hence the same-residual factor-exchange divisibility from TPC-57 is not available for this sum.

Proof. Expanding the square in (152) gives

$$|H_i|^2 = \sum_s |A_{i,s}|^2 + 2 \operatorname{Re} \sum_{s < t} A_{i,s} \overline{A_{i,t}}.$$

Summing over i proves (156). Substitution of (154) proves (157). The index condition $s < t$ implies (158). In contrast, the factor-exchange identity begins with equality of the two residual products, so its hypothesis fails here. $\square$

The absence of the factor-exchange divisibility does not say that (157) is large or negative. It says that a new signed estimate is required. An absolute pair count is generally too coarse because the desired information is one-sided spectral control.

To state the exact target, assume the high coefficient map is nonzero and let

$$R_H := T_H^* T_H, \quad N_H := T_H^* C_{\mu,H}^* C_{\mu,H} T_H, \quad (159)$$

so that $\operatorname{Ran} R_H \neq \{0\}$, and on $\operatorname{Ran} R_H$ define

$$\Theta_H := R_H^{\dagger/2} (N_H - R_H) R_H^{\dagger/2}. \quad (160)$$

Corollary 7.4 (Exact negative-spectrum gate). *For $0 < \delta \leq 1$, the uniform estimate*

$$\mathcal{N}_H(\zeta) \geq \delta \mathcal{G}_{\text{res}}(\zeta) \quad \text{for every } \zeta \quad (161)$$

is equivalent to

$$\lambda_{\min}(\Theta_H) \geq -1 + \delta. \quad (162)$$

Equivalently, the signed exchange form must satisfy

$$2 \operatorname{Re} \sum_i \sum_{s < t} A_{i,s} \overline{A_{i,t}} \geq -(1 - \delta) \mathcal{G}_{\text{res}} \quad (163)$$

for every coefficient vector. Only the negative spectrum must be controlled; no upper bound for the positive exchange is required.

Proof. On $\operatorname{Ran} R_H$, the generalized native Gram is

$$R_H^{\dagger/2} N_H R_H^{\dagger/2} = I + \Theta_H.$$

The Rayleigh–Ritz principle makes (161) equivalent to (162). The identity in [proposition 7.3](#) gives the equivalent form (163). $\square$

For a distinguished coefficient vector, the weakest pointwise target is just (163) for that vector. Proving it uniformly over the bounded row-coefficient cube is stronger and is exactly the positivity of the restricted constant α_H .

7.4 The strict endpoint ledger

Let $\mathcal{D}_{\text{par}}$ be the raw parent atomic energy on one declared carrier. The endpoint very-high cutoff forces the smaller cofactor range (20), so first suppose that a lower relocation transfer from the inherited parent packet into a compatible cofactor band has been proved. Suppose further that selection, cofactor-block occupancy, terminal activation, and the undeleted TPC–57 singleton bridge have given

$$\mathcal{G}_{\text{res}} \geq X^{-(\lambda_{\text{reloc}} + \lambda_{\text{sel}} + \lambda_{\text{blk}} + \lambda_{\text{term}} + \lambda_{\text{sing}}) + o(1)} \mathcal{D}_{\text{par}}. \quad (164)$$

Here $\lambda_{\text{reloc}} \geq 0$ records the parent-to-compatible-band loss and is not absorbed silently into λ_{blk} . This is a conditional input: TPC–57 neither proves this relocation nor proves the remaining occupancy and activation transfers asymptotically. On the inherited $R \asymp J$ carrier, the fixed- η very-high face is in fact empty for all sufficiently large X .

Assume also

$$\alpha_H \geq X^{-\lambda_H + o(1)}, \quad \delta_{\text{aff}} \geq X^{-\lambda_{\text{aff}} + o(1)}, \quad (165)$$

with nonnegative exponents. Boundary restriction and tail reconstruction, if available as lower transfers, are assigned losses λ_{bdry} and λ_{tail} .

Proposition 7.5 (Conditional endpoint propagation). *Under the preceding lower-transfer hypotheses, the prospective final physical quantity satisfies*

$$\mathcal{N}_{\text{final}} \geq X^{-\Lambda_{\text{tot}} + o(1)} \mathcal{D}_{\text{par}}, \quad (166)$$

where

$$\boxed{\Lambda_{\text{tot}} := \lambda_{\text{reloc}} + \lambda_{\text{sel}} + \lambda_{\text{blk}} + \lambda_{\text{term}} + \lambda_{\text{sing}} + \lambda_H + \lambda_{\text{aff}} + \lambda_{\text{bdry}} + \lambda_{\text{tail}}.} \quad (167)$$

Thus the safe endpoint requirement is

$$\boxed{\Lambda_{\text{tot}} < \frac{1}{400}.} \quad (168)$$

At equality, a separate closed constant-level comparison and quantified control of every $o(1)$ term are required; equality of formal exponents alone does not absorb an error of size $X^{-1/400 + o(1)} \mathcal{D}_{\text{par}}$.

Proof. Apply [theorem 6.4](#) to (164), then multiply the two assumed Gram transfers and the downstream boundary and tail transfers. Polynomial loss exponents add, giving (167). If $\Lambda_{\text{tot}} < 1/400$, the main lower scale dominates an additive $X^{-1/400+o(1)}\mathcal{D}_{\text{par}}$ error by a fixed power. When the two exponents are equal, the unspecified constants and $o(1)$ terms decide the sign, so exponent bookkeeping alone is insufficient. $\square$

For the direct already-reassembled fixed- h_0 route, no frozen-shift localization or projective-reassembly exponent is inserted. If another route passes through a completed frozen family or a signed projective decomposition, its localization and lower-reassembly losses must be restored in (167). It is convenient to record

$$\lambda_{\text{Gram}} := \lambda_H + \lambda_{\text{aff}}, \quad (169)$$

but the two terms must remain separately auditable because their kernels are independent.

8 Audit, claim boundary, and the next Gram gate

The point of the preceding construction is to locate, rather than conceal, the two comparisons that are still missing. With the final sign convention of [proposition 6.1](#),

$$z_H = -\sum_p b, \quad h = C_{\mu,H} z_H = -\sum_p v_p, \quad A = U_{\mu} z_H, \quad C_1 A = h. \quad (170)$$

Thus the physical affine column is $v_0 + h$. The first missing constant controls the erasure $z_H \mapsto h$, and the second controls the sum $(v_0, h) \mapsto v_0 + h$. They have independent kernels, so they must not be combined into one unnamed “Gram loss.”

8.1 Six-point physical audit

The native-affine bridge passes the following bookkeeping audit. Passing the audit certifies that the operators act on the intended physical object; it does not certify positivity of either restricted singular value.

- (A1) *Literal coefficients.* The terminal amplitude is the inherited physical coefficient $x_u = c_u \zeta_{m(u)}$, including its output factor, source shape, masks, profiles, cofactor sign, and terminal multiplier. In the TPC–56–57 common gauge, x_u already contains the factor $\mu(s(u))$. Equation (170) therefore uses the unsigned second grouping $C_1 A$; no second Möbius sign is inserted.
- (A2) *Fixed physical shift.* Every terminal atom satisfies $p_u r_u - m(u) j_u = h_0$ for the one declared nonzero physical shift. The identities in [propositions 6.1](#) and [7.3](#) are pointwise at this h_0 . No intercept average, frozen-shift family, or projective reassembly is substituted for the physical column.
- (A3) *Physical atomic normalization.* The terminal diagonal energy, the first grouped residual energy, and the native high energy use the inherited raw counting-measure atoms. Repeated cofactor labels remain distinct terminal atoms until the declared grouping, and the terminal multiplier is not moved into a renormalized input norm. Consequently the two grouping maps compare actual physical energies rather than support cardinalities.
- (A4) *Canonical representation and minimality scope.* The first grouping uses the full residual key $q(u) = (i, s(u))$; the second grouping erases only s . The restricted space $E_H = \text{Ran } T_H$ and the feasible affine pair space $\mathcal{F}_{\text{aff}}$ are the canonical images of one declared coefficient map. Moore–Penrose reductions are restricted to the active Gram range, so null coefficient directions are not counted as physical generalized eigenvectors.

- (A5) *Actual active support.* The baseline carrier used to define T_H is fixed independently of the row coefficient vector. Only factors that vanish identically on that baseline are removed. Coefficient-dependent zeros may shrink a distinguished support, but they do not redefine E_H , its rank, or the uniform constants. The high residual identification also uses the same $p > y$ cutoff on both the Walsh and terminal sides. At the endpoint this cutoff forces the smaller cofactor range (20); occupancy of that range is not inferred from the old $R \asymp J$ block.
- (A6) *Endpoint budget.* The two new losses are recorded separately as λ_H and λ_{aff} . Together with the separately recorded compatible-band relocation, selection, cofactor-block, terminal, singleton, boundary, and tail losses, they enter the exact ledger (167). The safe condition is the strict inequality $\Lambda_{\text{tot}} < 1/400$; equality is not accepted without a constant-level comparison and quantified remainder.

8.2 Claim table

Statement	Status
Signed residual-label erasure spectrum and bright/dark decomposition	Exact finite-dimensional theorem on the declared ambient fibers (L0).
Support-only native lower bound	Sharply false whenever one native output contains at least two active residual labels (L0 no-go).
Restricted native constant α_H	For nonzero $E_H = \text{Ran } T_H$, an exact least-eigenvalue and kernel characterization, with one-way rank/dimension kill tests (L0). Positivity and endpoint size are not proved.
Affine constant δ_{aff}	Exact restricted singular-value and anti-diagonal kernel characterizations on nonzero $\mathcal{F}_{\text{aff}}$ (L0). Positivity and endpoint size are not proved.
Two-stage affine transfer	Exact implication $\mathcal{N}_{\text{aff}} \geq \delta_{\text{aff}} \alpha_H \ z_H\ ^2$ for vectors in the declared feasible images (L0).
Fixed- h_0 terminal/Walsh/native identification	Exact coordinatewise interface under the inherited injection, cutoff, and common-carrier hypotheses (L1).
TPC-56 adaptive selected sector	Valid for its selected pre-terminal object, but it does not transport to the undeleted residual or native vector (L0 nontransport theorem).
TPC-57 terminal singleton bridge	Controls only the first grouping $p \mapsto (i, s)$, conditionally on its own occupancy hypotheses and after a compatible smaller cofactor band has been activated; it does not control the second grouping $(i, s) \mapsto i$ (L0/L1 boundary).
Native signed exchange identity	Exact expansion over unequal residual products at one fixed h_0 (L1). The preceding same-residual factor-exchange estimate does not apply to these pairs.
Uniform native high lower square	On nonzero $\text{Ran } R_H$, equivalent to the negative-spectrum bound $\lambda_{\min}(\Theta_H) > -1$; not proved.
Full native affine lower square	Requires both a native high lower square and independent base-high noncancellation; not proved.
Endpoint 1/400 comparison	Conditional exponent propagation only. None of the missing positive constants is entered as zero-cost.
Parity improvement or twin-prime consequence	Not claimed.

The level terminology is deliberately conservative. L0 comprises exact finite-dimensional identities, spectra, restricted minimax laws, and sharp counterexamples. L1 is the exact specialization of those maps to the literal fixed- h_0 carrier. An L2 advance would require a quantitative arithmetic estimate proving a sufficiently strong lower bound for the distinguished

physical vector or for the whole canonical coefficient image, together with every upstream and downstream transfer in (167). No theorem in this paper is classified as L2.

8.3 The next gates and their kill tests

The remaining work begins with one support door and then separates into two spectral doors. Compatible-band activation must be tested before either spectral constant: a Gram theorem on an empty inherited block carries no endpoint energy. After activation, the native door precedes the affine door because an affine estimate cannot reconstruct a high component already lost in residual-label erasure.

Door A. *Compatible-band relocation and occupancy.* With $y = X^{267/400+\eta}$, replace the incompatible old block $R \asymp J$ by dyadic blocks contained in

$$r \leq X^{133/400-\eta+o(1)}.$$

Prove a lower transfer from the inherited parent diagonal into these blocks, retain one fixed h_0 , and then verify terminal activation and the hypotheses of the TPC-57 undeleted singleton bridge. The relocation loss λ_{reloc} and subsequent block loss λ_{blk} remain distinct.

The immediate kill test is asymptotic emptiness or a best possible parent-to-band exponent already exhausting the $1/400$ slack. In either case the endpoint route stops before any Gram computation.

Door B. *Literal restricted-Gram assembly.* Assemble the pair

$$R_H = T_H^* T_H, \quad N_H = T_H^* C_{\mu,H}^* C_{\mu,H} T_H$$

from the actual fixed-carrier coefficients, after quotienting only $\text{Ker } T_H$. The first target is a certified census of the negative spectrum of Θ_H , not merely a floating-point plot. Exact sparse elimination, interval eigenvalue bounds, and blockwise Schur-complement certificates may be used, provided every block keeps the literal normalization and active support.

The immediate kill test is

$$E_H \cap \text{Ker } C_{\mu,H} \neq \{0\}.$$

If it holds on the actual image, then $\alpha_H = 0$ and the coefficient-uniform native door is closed. A positive numerical eigenvalue at finitely many scales is scouting evidence only; a proof requires a scale-uniform lower certificate with an explicit loss exponent.

Door C. *Distinguished signed exchange.* If the uniform image contains a dark vector, retain the one inherited coefficient vector and estimate the weakest exact target

$$2\Re \sum_i \sum_{s < t} A_{i,s} \overline{A_{i,t}} \geq -(1 - \delta_H) \sum_{i,s} |A_{i,s}|^2 \quad (171)$$

with $\delta_H \geq X^{-\lambda_H+o(1)}$. Unlike the TPC-57 same-residual collision system, the pairs here satisfy $s(u) \neq s(v)$. Any arithmetic proof must therefore expose a new unequal-residual-product relation rather than reusing singleton occupancy under a renamed key.

This door is closed for the endpoint if literal examples force δ_H to decay faster than the residual endpoint slack. Such a result would be a genuine negative theorem about this route, not a claim of cancellation for unrelated coefficient families.

Door D. *Base-high affine transversality.* Conditional on surviving Door B or C, assemble the feasible pair image $\mathcal{F}_{\text{aff}}$ and test

$$\mathcal{F}_{\text{aff}} \cap \{(g, -g) : g \in \mathcal{H}_{\text{out}}\} = \{0\}. \quad (172)$$

A nonzero intersection proves $\delta_{\text{aff}} = 0$ for the uniform route. Otherwise the target is a quantitative principal-angle lower bound $\delta_{\text{aff}} \geq X^{-\lambda_{\text{aff}}+o(1)}$. This analysis must retain the joint dependence of v_0 and h on the same coefficient vector; treating them as independent ambient vectors can only recover the sharp no-go of [theorem 6.5](#).

Door E. *Distinguished affine ratio.* If uniform affine transversality fails but the physical coefficient vector avoids the anti-diagonal, study the exact pointwise ratio [\(149\)](#). As at the native gate, a distinguished estimate is weaker than a coefficient-uniform theorem and must be labeled accordingly. Its loss exponent still enters the same strict endpoint ledger.

Door F. *Budget stop and route closure.* At each scale record $\lambda_{\text{Gram}} = \lambda_H + \lambda_{\text{aff}}$ separately from the five upstream and two downstream losses. If the best rigorous lower exponents force $\Lambda_{\text{tot}} \geq 1/400$, the current endpoint route is not certified. At equality, stop unless all constants and remainders are closed. If either exact kernel test is triggered, record the corresponding negative result and return to a different physical carrier or decomposition; do not replace the failed undeleted vector by the adaptive TPC-56 sector.

The most economical next paper is therefore a compatible-band census at the literal fixed shift: determine whether the smaller r -range forced by $p > y$ carries a quantitatively useful part of the parent energy. Only on a surviving nonzero carrier should the subsequent paper construct R_H , N_H , and the affine pair Gram and perform exact rank and kernel tests. This order permits a positive theorem or a sharp route-closure theorem without performing spectral analysis on an empty face.

Conclusion. The missing native bridge is now an exact signed alias problem. TPC-57 can, subject to its own unresolved arithmetic hypotheses, deliver energy after the first grouping, but residual-label erasure introduces the new unequal-residual exchange form. Its optimal restricted constant is α_H . Even if that gate is positive, completion of the physical column encounters the independent affine constant δ_{aff} . Both constants have exact spectral descriptions and sharp independent zero mechanisms; neither is proved positive on the endpoint carrier. The paper therefore closes an L1 interface and exposes one upstream activation gate and the next two auditable spectral gates, while making no parity improvement and no twin-prime claim.

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